

Master Documentation __Generated 2025-10-29 05:31__

Total files: 224 Total size: 43.2 MB

File Types

{'py': 42, 'txt': 5, 'md': 4, 'pdf': 21, 'noext': 3, 'pkl': 7, 'ipynb': 22, 'yaml': 3, 'json': 1, 'jpg': 32, 'xlsx': 4, 'png': 44, 'csv': 28, 'joblib': 5, 'db': 1, 'toml': 1, 'gitkeep': 1}

Repository Tree

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### .devcontainer - .devcontainer/devcontainer.json (1.5 KB) ###
.git - .git/FETCH_HEAD (139.0 B) - .git/HEAD (21.0 B) - .git/config
(428.0 B) - .git/config.worktree (83.0 B) - .git/description (73.0 B) -
.git/hooks/applypatch-msg.sample (478.0 B) - .git/hooks/commit-msg.sample
(896.0 B) - .git/hooks/fsmonitor-watchman.sample (4.6 KB) - .git/hooks/post-update.sample
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.git/logs/HEAD (218.0 B) - .git/logs/refs/heads/main (227.0 B) -
.git/logs/refs/remotes/origin/main (326.0 B) - .git/objects/pack/pack-1d3d1df4b8e4eb4e9a584bcc9
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(20.1 MB) - .git/objects/pack/pack-1d3d1df4b8e4eb4e9a584bcc98c74084192761bb.rev
(1.0 KB) - .git/refs/heads/main (41.0 B) - .git/refs/remotes/origin/main
(41.0 B) - .git/shallow (41.0 B) ### .github - .github/workflows/autodoc.yml
(5.1 KB) - .github/workflows/gen-tree.yml (949.0 B) - .github/workflows/generate-pdf.yml
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- cb\_dashboard\_artifacts/data/benchmarking\_data\_fakergen.xlsx
(12.0 KB) - cb\_dashboard\_artifacts/data/employee\_compensation\_fakergen.xlsx
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KB) - cb\_dashboard\_artifacts/exports/Cb\_Report\_consolidated.pdf
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 data/xgb_top_features.csv (1.4 KB) ### faker_data - faker_data/FakerGen_CB.ipynb
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 (176.5 KB) - reports/Attrition_ModelExplainability_52d61660976d1481ab051617939779a2.pdf
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 reports/Attrition_SHAP_Explainability_V1_883d0c6ff33ddb40c5459b894a884b62.pdf
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```
## Code & Notebook Overview
```

```
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/generate\_autodoc.py
import pathlib, os, datetime, subprocess, collections, nbformatROOT = pathlib.Path('.').resol
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/cb\_dashboard.py
\# =====\# cb\_dashboard.py - Compensation \& Benefits Dash
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/app.py
\# app.py -- HR Attrition Prediction Dashboard (Final Polished Version)\# Save this file as
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/models/Attrition\_AdvancedModels
\# Project 5: Advanced Attrition Models (Tuned Random Forest + XGBoost)
```

In this notebook, we benchmark **and** enhance predictive models **for** employee attrition.

- Start **with** tuned ****Random Forest****
- Introduce ****XGBoost**** (gradient boosting)
- Compare performance vs Logistic Regression

```
**Goal:**
```

Improve predictive accuracy **while** balancing interpretability **and** business storytelling.

```
\# -----
\# Imports
\# -----
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import joblib
```

```
from sklearn.model\_sele
```

```
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/scripts/export\_pdf.py
import osimport subprocessimport hashlibfrom pathlib import Pathfrom nbconvert import HTMLEx
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/scripts/gen\_tree.py
import osfrom pathlib import Pathdef generate\_tree(start\_path=".", prefix="", max\_items=3
```

```

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/faker\_data/FakerGen\_CB.ipynb
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main,
\# -----
\# Step 0 - Clone hr-tech-portfolio (with PAT)
\# -----
import os
from getpass import getpass

\# Ask for your GitHub Personal Access Token
token = getpass(" Enter your GitHub token: ")

\# Clean any previous clone
os.system("rm -rf hr-tech-portfolio")

\# Clone with token authentication
os.system(f"git clone

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/helper\_scripts/sql\_utils.py
import sqlite3, pandas as pd, osDB\_PATH = os.path.abspath("hr\_dataset.db")def run\_query(s
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/helper\_scripts/hr\_db.py
import pandas as pd, sqlite3, osDB\_PATH = os.path.abspath("hr\_dataset.db")CSV\_PATH = os.p
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/HR\_DataForge.
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main,
!pip install faker pandas numpy

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/app.py
\# =====\# app.py - People Analytics Dashboard (v3.9
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils/fusion\_
\# =====\# utils/fusion\_report\_builder.py - v2.3 /
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils/ui\_styl
\# =====\# utils/ui\_styling.py - v1.0 | Unified Side
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils/pdf\_exp
\# =====\# utils/pdf\_explainer\_builder.py - v5.8 /
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils/chart\_s
\# =====\# utils/chart\_saver.py - v4.4 | ColorSafe
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils/pdf\_aut
\# =====\# utils/pdf\_auto\_exporter.py - v3.7 | Fil
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils/persist

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\# =====\# utils/persistence\_chatbot.py - v2.6 / Re
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis/fix\_help
\# =====\# utils/fix\_helper.py - v3.0.3 / Final Syn
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis/template
import streamlit as stimport pandas as pd\# --- Unified Styling for All Download Buttons ---
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis/uploader
\# =====\# utils\_consolidated/uploader\_consolidated
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis/\_init
"""utils package initializer-----Ensures core utility modules load clean
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis/pdf\_help
\# =====\# utils/pdf\_helper.py - v5.2 / Executive S
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/pages/compensa
import streamlit as stst.set\_page\_config(page\_title="Compensation Analytics", layout="wide")
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/pages/workforce
import streamlit as stst.set\_page\_config(page\_title="Workforce Analytics", layout="wide")
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/pages/engagement
import streamlit as stst.set\_page\_config(page\_title="Engagement Analytics", layout="wide")
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/pages/performance
import streamlit as stst.set\_page\_config(page\_title="Performance Analytics", layout="wide")
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/pages/consolidated
\# =====\# pages/consolidated.py - v6.8 / Auto Job De
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/pages/attrition
import streamlit as stst.set\_page\_config(page\_title="Attrition Analytics", layout="wide")
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis\_consolidated
\# =====\# utils\_consolidated/deck\_state\_tracker.py
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis\_consolidated
\# =====\# utils\_consolidated/chart\_consolidated\_state\_tracker.py
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis\_consolidated
\# utils\_consolidated/insights\_helper.py"""Small helper that consolidates insights lists
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utis\_consolidated
\# utils\_consolidated/constants.py\# Theme + layout constants for consolidated PDF builder

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### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils\_consolidated
\# =====\# utils\_consolidated/pdf\_helper\_consolidated
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils\_consolidated
\# =====\# utils\_consolidated/pdf\_merger.py - v7.5
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils\_consolidated
\# utils\_consolidated/uploader\_consolidated\_helper.py\# v1.3 - uploader that is mobile-friendly
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils\_consolidated
\# utils\_consolidated/toc\_helper.py""Simple TOC helper: builds a table-friendly TOC array
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils\_consolidated
\# =====\# utils\_consolidated/pdf\_consolidated\_helper.py
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/utils\_consolidated
""utils\_consolidated package initializer-----Ensures all
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/modules/consolidated
\# =====\# modules/consolidated\_module.py - v7.0 | Ensures all
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/modules/workforce
\# =====\# modules/workforce\_module.py - v3.0.1 | Ensures all
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/modules/engagement
\# =====\# modules/engagement\_module.py - v3.0.1 | Ensures all
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/modules/consolidated
\# =====\# consolidated\_module\_diagnostic.py - Minimum
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/modules/attrition
\# =====\# modules/attrition\_module.py - v3.2 | Ensures all
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/modules/performance
\# =====\# modules/performance\_module.py - v3.0.1 | Ensures all
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people\_analytics/modules/compensation
\# =====\# modules/compensation\_module.py - v3.0.1 | Ensures all
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\_SQL\_Integration
\# Project 8: SQL + ML Integration

```

****Objective:****

Combine ****SQL querying power**** with ****Machine Learning models**** to analyze attrition risk. This project demonstrates how HR teams can query their employee database directly and run pr

****Why It Matters:****

- HR data often lives **in** databases (HRIS, payroll systems).
 - Analysts should be able to run queries **and** pipe results into ML models.
 - This integration makes predictive attrition analytics more practical **in** enterprise context
- \#\# SQL + ML Integration*

Thi

```
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Day4-AttritionRiskAnal  
\# Attrition Risk Analyzer (v2.0)
```

\#\# Objective

Analyze IBM HR Attrition dataset to identify attrition patterns **and** create a simple attrition

\# Import required libraries

```
import pandas as pd
```

```
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\_PredictiveM
```

\# Predictive Model - Attrition (Logistic Regression)_V3 - Fine Tuning

\# -----

\# Imports

\# -----

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import os
```

```
from sklearn.model\_selection import train\_test\_split, cross\_val\_score, GridSearchCV
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.linear\_model import LogisticRegression
```

```
from sklearn.metrics import (
```

```
    accuracy\_score, classification\_report, confusion\_matrix, roc\_auc\_score
```

```
)
```

```
import joblib
```

```
sns.set(style="whitegrid")
```

```
pd.set\_option('display.max\_c
```

```
#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\_ModelCompar
```

\# Project 4: Attrition Model Comparison (Logistic vs Random Forest)

In this notebook, we extend the attrition prediction by comparing ****Logistic Regression**** (1

****Goal:****

Check **if** tree-based models improve predictive performance **and** what new insights they provide

```

\# -----
\# Imports
\# -----
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation\_Analytics\_

\# -----
\# Step 0 - Clone Repo with PAT Auth
\# -----

import os
from getpass import getpass

\# If repo already exists in Colab, remove it \(fresh start\)
!rm -rf hr-tech-portfolio

\# Ask for PAT token securely
token = getpass\("Enter your GitHub PAT: "\)

\# Clone using HTTPS + token auth
!git clone https://

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\\_AdvancedModeling\\_
\# Project 5: Advanced Attrition Models \(Tuned Random Forest + XGBoost\)

In this notebook, we benchmark and enhance predictive models for employee attrition.
- Start with tuned \*\*Random Forest\*\*
- Introduce \*\*XGBoost\*\* \(gradient boosting\)
- Compare performance vs Logistic Regression

\*\*Goal:\*\*
Improve predictive accuracy while balancing interpretability and business storytelling.
\# -----
\# Imports
\# -----
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

```

```

import seaborn as sns
import os
import joblib

from sklearn.model\_sele

### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\_PredictiveM
\# Predictive Model - Attrition (Logistic Regression)
**Objective:** Train a logictic regression model to predict employee attrition (Yes/No) usin
\# Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")

from sklearn.model\_selection import train\_test\_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear\_model import LogisticRegression
from sklearn.metrics import accuracy\_score, classification\_report, confusion\_matrix, roc
import os

\# Display settings
pd.set\_option('displ

### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Day4-AttritionRiskAnal

import pandas as pd
df= pd.read\_csv("WA\_Fn-UseC\_HR-Employee-Attrition.csv")
print("Shape of Data:",df.shape) \#Explore Dataset
print(df.head())
print(df['Attrition'].value\_counts()) \#Quick Summary of Attrition
\#Group by Job Role
job\_attrition=df.groupby(['JobRole','Attrition']).size()
job\_attrition=job\_attrition.unstack().fillna(0)
print(job\_attrition)
\#Simple Attrition Risk Flag
df['AttritionRisk']=df['Attrition'].apply(lambda x:1 if x=="Yes" else 0)
\#Save Processed File
df.to\_csv("processed\_hr\_data.csv",index=False)
print("Attrition Risk Analyzer v0.1 ready ")

### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation\_Analytic

<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/
from getpass import getpass

\# GitHub authentication

```

```

token = getpass(" Enter your GitHub token: ")

repo\_url = f"https://AMBOT-pixel96:\{token\}@github.com/AMBOT-pixel96/hr-tech-portfolio.git"

!git clone \{repo\_url\}
\%cd hr-tech-portfolio

### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/
Attrition\_PredictiveModel.ipynb

\# Predictive Model - Attrition (Logistic Regression)
**Objective:** Train a logistic regression model to predict employee attrition (Yes/No) using
\# Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")

from sklearn.model\_selection import train\_test\_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear\_model import LogisticRegression
from sklearn.metrics import accuracy\_score, classification\_report, confusion\_matrix, roc\_curve
import joblib
import sqlite3

\# displ

### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\_ModelExplainability.ipynb
\# Project 6: Model Explainability with SHAP

In this notebook, we apply **SHAP (SHapley Additive exPlanations)** to interpret machine learning models.

**Goals:**
- Understand which features drive attrition globally across employees
- Explain individual predictions for specific employees
- Provide visuals that HR leaders can use to trust model decisions
\# 1 Imports

### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\_Streamlit\_App.ipynb
> (Could not preview: Notebook does not appear to be JSON: '# Save training
columns for alignment\n...') ### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition\_Streamlit\_App.ipynb
\# Project 9: Attrition Explainability with SHAP

**Objective:**
Enhance the HR Attrition prediction models by adding explainability using SHAP (SHapley Additive
This notebook will:

```

1. Load saved models **from** Project 7 (Deployment).
2. Apply SHAP to explain feature contributions.
3. Export explainability visuals **for** reports **and** dashboards.

****Artifacts generated:****

- SHAP summary plots
- SHAP bar plots
- SHAP force plots (employee-level explanations)
- Stored outputs **in** `~/reports/09/_explainability/``

```
import pandas as pd
import shap
import joblib
import os
fr
```

```
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation\_Analytics
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/`
\# Project 10: Compensation Analytics (v1.0)
```

****Objective:****

Analyze synthetic HR Compensation dataset to extract key insights: pay distribution, gender

****Business Context:****

Compensation & Benefits (C&B) analytics helps HR leaders answer:

- Are we paying fairly across levels, genders, departments?
- How are bon

```
### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation\_Analytics
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/`
```

```
\# Clone repo + move into it
```

```
from getpass import getpass
token = getpass(" Enter your GitHub token: ")
```

```
repo\_url = f"https://AMBOT-pixel96:\{token\}@github.com/AMBOT-pixel96/hr-tech-portfolio.git"
!git clone \{repo\_url\}
\%cd hr-tech-portfolio
```

```
\# Import libraries
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/baby\_steps/Day2-Basic
\#1.Variables
name="Amlan"
age="29"
print("My name is",name," and I am",age," years old")
\#2.Simple Math
salary=1404000 \#Yearly INR
bonus=salary*.12
total=round(salary+bonus)
print("Total Annual Pay:",total)

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/baby\_steps/Day1-Hello
print("Hello Amlan - HR Tech Transformation ")

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/baby\_steps/Day3-DataT
\#1.Dictionaries
employee=\{
    "name": "Amlan",
    "role": "HR Tech",
    "salary": 1404000,
    "skills": ["PowerBI", "Oracle Fusion", "Python"]
\}
print(employee["name"],"earns",employee["salary"],"as a",employee["role"],"professional with
\#2.If-Else
attrition\_risk =0.7
if attrition\_risk > 0.5:
    print("High Risk - Action Needed")
else:
    print("Low Risk - All Good")

#### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/sidequests/HR\_Data\_Cleaning\_U
\# HR Data Cleaning Utilities (v1.0)

```

This notebook demonstrates how to ****simulate messy HR data**** and then build a cleaning pipeline. Data cleaning **is** a critical step **in** People Analytics - poor quality data = misleading insights.

```

\# Import libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import random

```